ICKII

Conference Presentation

A Neuromorphic Human Detection System Using Thermal and Event Cameras for Humanoid Robotics

Authors: Rui Shiota, Abderazek Ben Abdallah, Khanh N. Dang and Zhishang Wang

The School of Computer Science and Engineering, The University of Aizu, Aizu-Wakamatsu 965-8580, Fukushima, Japan

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Contents

Introduction

RESEARCH CONTEXT

Event-driven perception

Physical motion
Scene change

→

GENX320
Brightness events

→

Event-rate trace
windows

→

Motion decision
Motion / no motion

Research topic	Platform	Output
Motion detection	OpenMV + FPGA	Binary output
Current evidence	Recorded traces	Preliminary metrics

Motivation

WHY COUNT EVENTS

Property	Frame camera	Event camera
Output	Full frames	Brightness changes
Timing	Fixed rate	Asynchronous
Processing	Full-image analysis	Event counting

Event stream

x, y coordinates · polarity · time

windows

Event count

Output

Motion / no motion

Continuous motion check by event counting

Contribution

OBJECTIVE

Layer	Content	Evidence	Status
Proposal	Event-rate rule	Algorithm definition	Defined
Evaluation	Basic scenarios	29,366 windows	Completed
Hardware	UART RX → LED	Basys 3 check	Verified
System	FPGA metrics / power	End-to-end test	Pending

Output **MOTION / NO MOTION**

FPGA check **UART reception**

System and Evidence Paths

METHOD

GENX320
x, y coordinates
polarity · timestamp

→

H7 Plus
20 ms event count

→

UART data
Event count
Motion flag

→

Basys 3
Threshold · 3 windows

→

Output
Motion / no
motion

Recorded data

Basic scenarios → Metrics

Challenge scenarios → Behavior

Hardware check

UART → FPGA → LED

Offline ≠ FPGA

Detection Rule

METHOD

1. Count — 20 ms → **2. Rate** — Events / second → **3. Compare** — > 10,000/s → **4. Confirm** — 3 windows

Parameter	Value	Role
Window	20 ms	Count
Threshold	10,000/s	Decide
Filter	3 windows	Confirm

Temporal filter

HIGH → HIGH → HIGH → MOTION

3 × 20 ms = 60 ms

Confirmation delay: 60 ms

Evaluation Data

EVALUATION

Data	Scenarios	Use
Basic scenarios	Walk · wave · sit–stand · empty	Accuracy / recall / F1
Challenge scenarios	Entry/exit · chair · dropped object	Behavior check

Basic scenarios **20 files · 29,366 windows · metrics**

Challenge scenarios **15 files · behavior check only**

Evaluation Metrics

EVALUATION

Ground truth	Rule output	Count
MOTION	MOTION / NO MOTION	TP / FN
NO MOTION	MOTION / NO MOTION	FP / TN

Labels

MOTION Walk · wave · sitting–standing

NO MOTION Empty

Accuracy

(TP + TN) / All

Recall

TP / (TP + FN)

Precision

TP / (TP + FP)

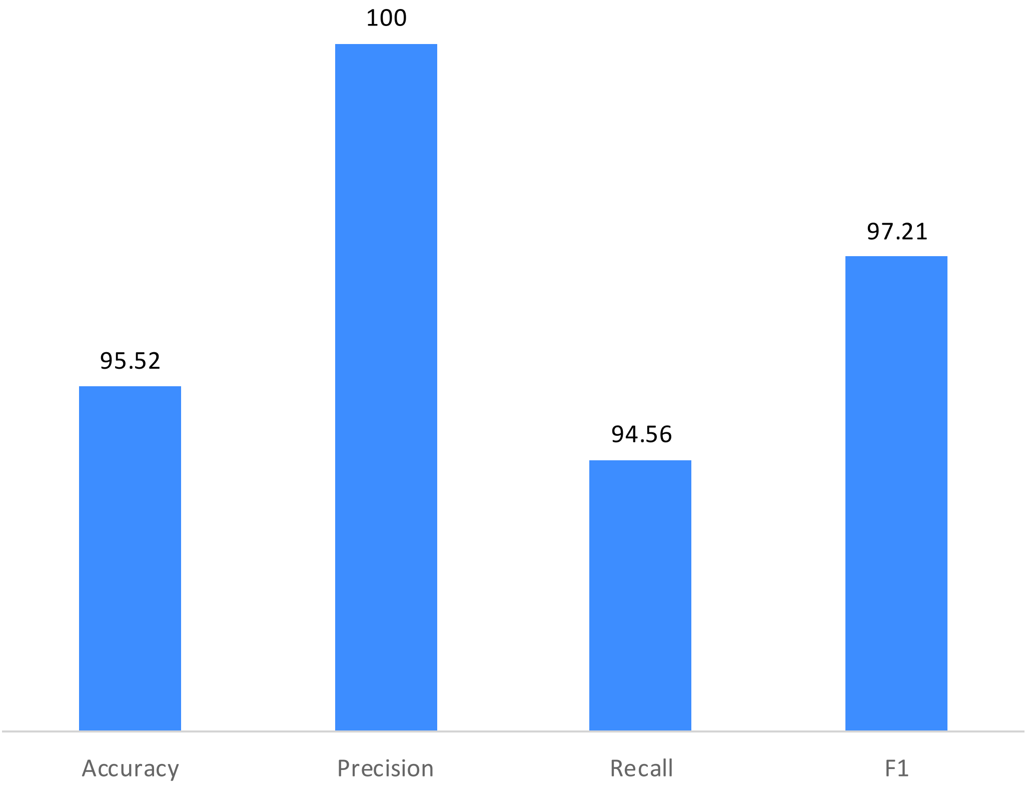
F1

P–R balance

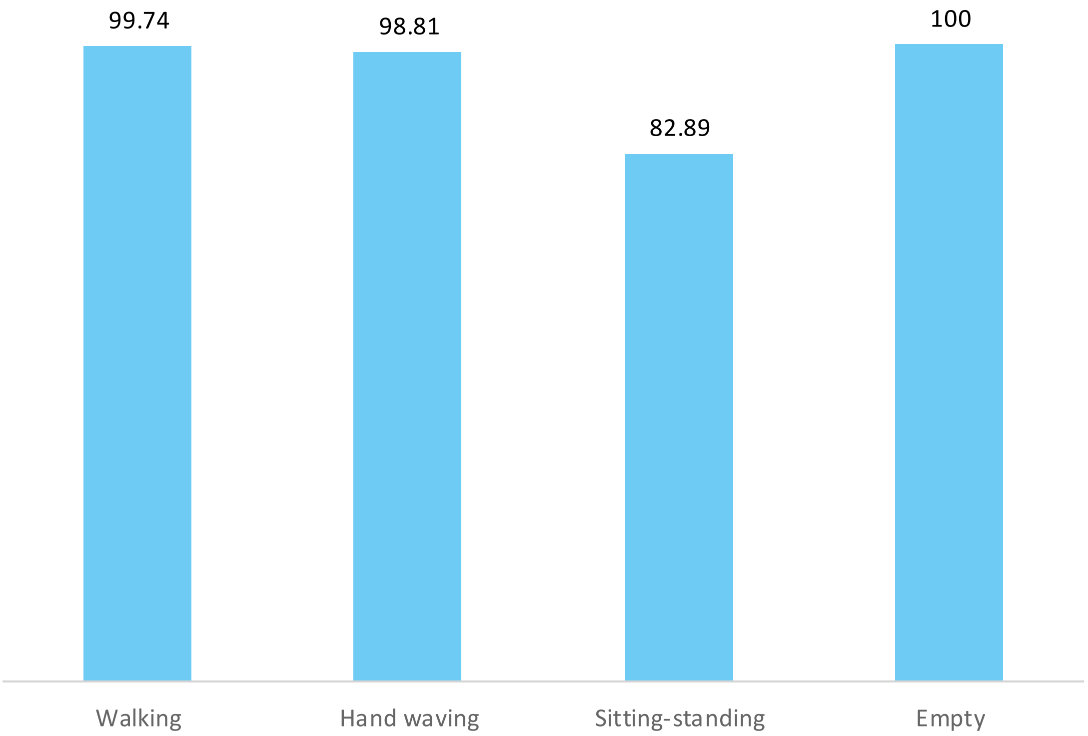
Unit: 20 ms window

Basic Scenario Results

RESULTS



Overall metrics



Correct decision rate by scenario

Discussion

SCOPE

Current evidence

- ✓ Basic scenarios
- ✓ Challenge scenarios
- ✓ Motion metrics
- ✓ UART reception

Validation roadmap

1. FPGA trace replay
2. Bit-level agreement
3. Latency + power
4. Event → thermal

Remaining work **FPGA replay · latency · power · sensor integration**

Discussion: required capability and deployment context

References

1. Shakir Sumit, D. Rambli, and S. Mirjalili, “Vision-Based Human Detection Techniques: A Descriptive Review,” IEEE Access, 2021. DOI: 10.1109/ACCESS.2021.3063028.

2. T. Kryjak, “Event-Based Vision on FPGAs - a Survey,” in Euromicro Symposium on Digital Systems Design, 2024. DOI: 10.1109/DSD64264.2024.00078.

3. J. Gupta and S. Sharma, “Towards Real Time Hardware Based Human and Object Detection: A Review,” in International Joint Conference on the Analysis of Images, Social Networks and Texts, 2022. DOI: 10.1109/AIST55798.2022.10064933.

4. B. Amirgaliyev, M. Mussabek, T. Rakhimzhanova, and A. Zhumadillayeva, “A Review of Machine Learning and Deep Learning Methods for Person Detection, Tracking and Identification, and Face Recognition with Applications,” Sensors, 2025, 25(5):1410. DOI: 10.3390/s25051410.

5. X. Wang et al., “HARDVS: Revisiting Human Activity Recognition with Dynamic Vision Sensors,” in AAAI Conference on Artificial Intelligence, 2022. DOI: 10.48550/arXiv.2211.09648.

6. X. Lin, M. Liu, and H. Chen, “Spike-HAR++: an energy-efficient and lightweight parallel spiking transformer for event-based human action recognition,” Frontiers in Computational Neuroscience, 2024. DOI: 10.3389/fncom.2024.1508297.

7. L. Li et al., “An Event-tailored State-Space Based Model for Pedestrian Detection,” in ACM Multimedia, 2025. DOI: 10.1145/3746027.3755380.

8. T. Trinh, P. C. Le Thien Vu, and P. T. Bao, “Comparing Mask R-CNN backbone architectures for human detection using thermal imaging,” International Journal of Electrical and Computer Engineering, 2024. DOI: 10.11591/ijece.v14i4.pp3962-3970.

9. R. Elgohary, A. K. Mahmoud, O. Khaled, S. Salah, and M. Badawi, “Enhanced Thermal Human Detection in Military Applications Using Deep Learning,” in 2024 International Telecommunications Conference (ITC-Egypt), 2024. DOI: 10.1109/ITCEgypt61547.2024.10620540.